

12th International ABINIT developer workshop

Sant Feliu de Guixols 8-11 June 2026

ABINIT project overview

Introduction to the workshop

X. Gonze
UCLouvain, Belgium

ABINIT software project

Ideas (1997) – pioneer :

- 1) Software applications for first-principles simulations are more and more sophisticated :
one needs a worldwide collaboration,
of specialized, complementary, groups
- 2) Linux software development : ‘free software’ model

Now (2026) :

Estimated > 1500 users worldwide
> 1 300 kLines of F90
> hundred python scripts
about 60 contributors to ABINITv8/v9/v10

ABINIT milestones

- Precursor : the Corning PW code (commercialized 1992-1995 by Biosym)
- 1997 : beginning of the ABINIT project
- Dec 2000 : release of ABINITv3 under the GNU General Public License (GPL)
- 2010 : launch of the old Forum (up to 2500 registered people) / 2021 : launch of the discourse Forum (now >1500 registered people).
- Nov 2002 – June 2026 : **12 international ABINIT developer workshops** (between 35 and 80 participants each). Split 2021-2022 => COVID/hackathon



... Now in Spain !

Impactful capabilities of ABINIT

‘Abinit 2025 : New capabilities for the predictive modelling of solids and nanomaterials’ J. Chem. Phys. 163, 164126 (2025)

M.J. Verstraete et al – 75 authors

Non-exhaustive list :

Dynamical Mean-Field Theory

GW + PAW, recently low-scaling GW + GWPT

Advanced capabilities in DFPT (strain perturbation, spin-magnetic field perturbation, flexoelectricity, dynamical quadrupoles, non-linear electric susceptibility, Raman intensity, natural optical activity ...)

cDFT and cDFPT

aTDEP

Electron-phonon : polarons, mobility calculations, zero-point renormalization

GPU porting ...

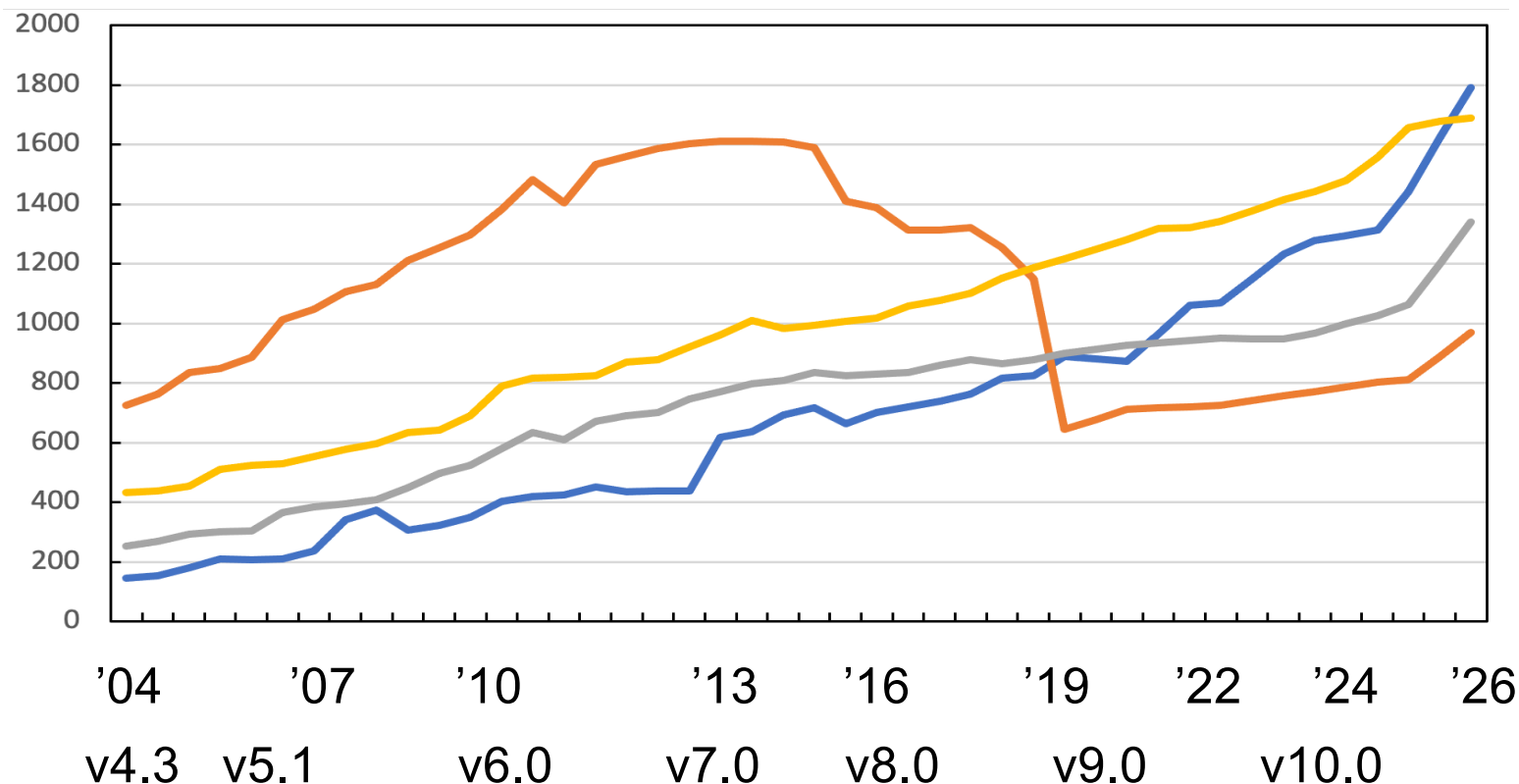
Also : Quality of documentation, especially tutorial

Abipy script gallery

Norm-conserving pseudos (pseudo-dojo) + JTH PAW atomic data



Time evolution : indicators



As of Feb 2026 (v10.6) :

1690	automatic tests
1340	kLines of F90
179.1	MB tar.gz package
970	F90 files

Also : AbiPy 0.9.0 gallery with >50 plotting scripts (python)
>40 tutorial

Bibliometry

Citations (scopus) of the 6 articles (<2021) co-authored by active developers
Total as of June 7, 2026 : 8502 citations ; 529 citations in 2025

Comput. Mat. Science 25, 478 (2002) – 16 authors
3033 citations ; 93 citations in 2025

Z. Kristallogr. 220, 558 (2005) – 20 authors
1250 citations ; 32 citations in 2025

Comp. Phys. Comm. 180, 2582 (2009) – 33 authors
2487 citations ; 112 citations in 2025

Comp. Phys. Comm. 205, 106 (2016) – 55 authors
776 citations ; 77 citations in 2025

Comp. Phys. Comm. 248, 107042 (2020) – 53 authors
658 citations ; 155 citations in 2025

J. Chem. Phys. 152, 124102 (2020) – 43 authors
298 citations ; 60 citations in 2025

Abilogix

A consortium to manage the development of ABINIT.

Consortium agreement : 1/10/2021 – 30/9/2031.

Institutional value !

Members : UCLouvain, ULiège, UNamur, CEA (1/10/23), UTQR (1/9/25).

In practice : meetings for coordination of the actions of permanent members incl. JMB, MG, HeXu

Composition of the board :

- Representatives of the 5 institutions
(XG, GMR, PhG, MV, LH, MT, BA)
- Elected representative of the ABINIT advisory board (JZ)
- Elected representative of the developers (to be elected HERE)
- Elected representative of the users (still to be organized)

Advisory board

The role of the ABINIT International Advisory Committee (IAC) is to help make decisions concerning broad policies of ABINIT implementation, use, and licensing, and to help coordinate international activities of the ABINIT community.

Examples of IAC activities include maintenance of the GPL licensing model within ABINIT, forging links to related coding projects, and coordinating ABINIT-sponsored Workshops and Schools.

D.C. Allan, G. Antonius, E. Bousquet, F. Bruneval, D. Caliste, M. Côté, Ph. Ghosez, X. Gonze, N. Holzwarth, F. Jollet, M. Mikami, S. Poncé, J. Rehr, G.-M. Rignanese, A. Romero (Chair), N. Spaldin, M. Stengel, M. Torrent, M. Verstraete, J. Zwanziger

Lots of things to do at ABIDEV26 !

Presentations by developers + “external” invitees

Discussion sessions / Hackathon

+ Advisory board / Migration / Abilogix dev. repr. / Forum Group picture

	Monday, June 8	Tuesday, June 9	Wednesday, June 10	Thursday, June 11
9h00	AI initiation	Discussions:	Hackathon(s)	Second-principles
		AI for coding,	AI "exercises"	and aTDAP
		documentation and	with ABINIT	
		input file generation		
10h30		Break	Break	(10h20) Break
11h00		Spin-phonon,	Feedback AI	(10h50) MBPT
		magnetism	(12h00) Review	
			previous hackathons	
13h00	Lunch	Lunch	Lunch	(12h30) Lunch
14h	Lunch	Spatial dispersion,	Data quality,	
14h30	Welcome	DFPT	code interfaces,	
	Scientific appetizers		LLM+RAG	
15h40	(15h50) Break	Break	Break	
16h10	(16h20) Optimisation,	Electron-phonon	Strong correlation,	
	algorithms, HPC		time dependence	
18h	(18h10) Posters	ABILOGIX-Forum	Advisory board	
19h30	Dinner	Dinner	Dinner	



ABIDEV 2026, June 8, 2026